

(12) **United States Patent**
Brajovic

(10) **Patent No.:** **US 12,409,952 B1**
(45) **Date of Patent:** **Sep. 9, 2025**

(54) **VEHICLES, SYSTEMS, AND METHODS FOR THE AUTONOMOUS INSPECTION OF OBJECTS**

G06T 2207/10028; G06T 2207/10032;
G06T 2207/20081; G06V 20/17; H04W
4/40; B64U 2101/26; B64U 2101/30;
B64U 2201/10
See application file for complete search history.

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 552 days.

(21) Appl. No.: **18/076,666**

(22) Filed: **Dec. 7, 2022**

Related U.S. Application Data

(60) Provisional application No. 63/287,598, filed on Dec. 9, 2021.

(51) **Int. Cl.**
B64F 5/60 (2017.01)
G01S 17/89 (2020.01)
G06T 7/00 (2017.01)
G06T 7/38 (2017.01)
G06V 20/17 (2022.01)
(Continued)

(52) **U.S. Cl.**
CPC **B64F 5/60** (2017.01); **G01S 17/89** (2013.01); **G06T 7/0004** (2013.01); **G06T 7/38** (2017.01); **G06V 20/17** (2022.01); **B64U 2101/26** (2023.01); **B64U 2101/30** (2023.01); **B64U 2201/10** (2023.01); **G06T 2207/10028** (2013.01); **G06T 2207/10032** (2013.01); **G06T 2207/20081** (2013.01)

(58) **Field of Classification Search**
CPC B64F 5/60; B64C 39/024; G01S 17/89; G06F 30/15; G06T 7/0004; G06T 7/38;

(56)

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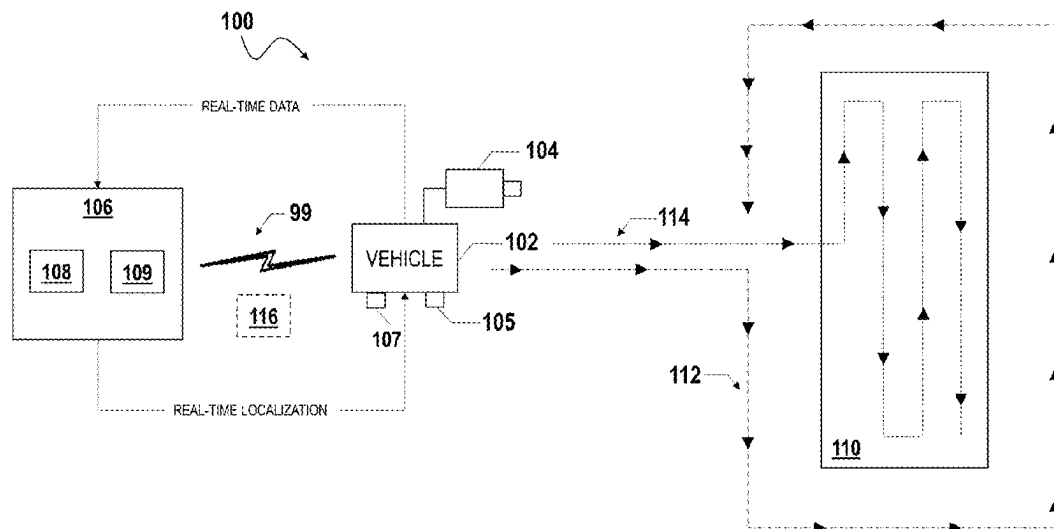
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(57)

ABSTRACT

A vehicle autonomously inspects an object within an environment. The vehicle includes an inspection sensor, a localization sensor, and a computing system. The localization sensor can detect three-dimensional surfaces within the environment. The inspection sensor can characterize the object. The computing system includes a memory configured to store a three-dimensional model of the object and predetermined inspection instructions. The computing system further includes a processor configured to register the model of the object within the environment based on information gathered by the localization sensor on an initial journey of the vehicle. The processor can also adjust the predetermined inspection instructions based on the information gathered by the localization sensor on an initial journey of the vehicle.

23 Claims, 8 Drawing Sheets



(51) **Int. Cl.***B64U 101/26* (2023.01)*B64U 101/30* (2023.01)(56) **References Cited**

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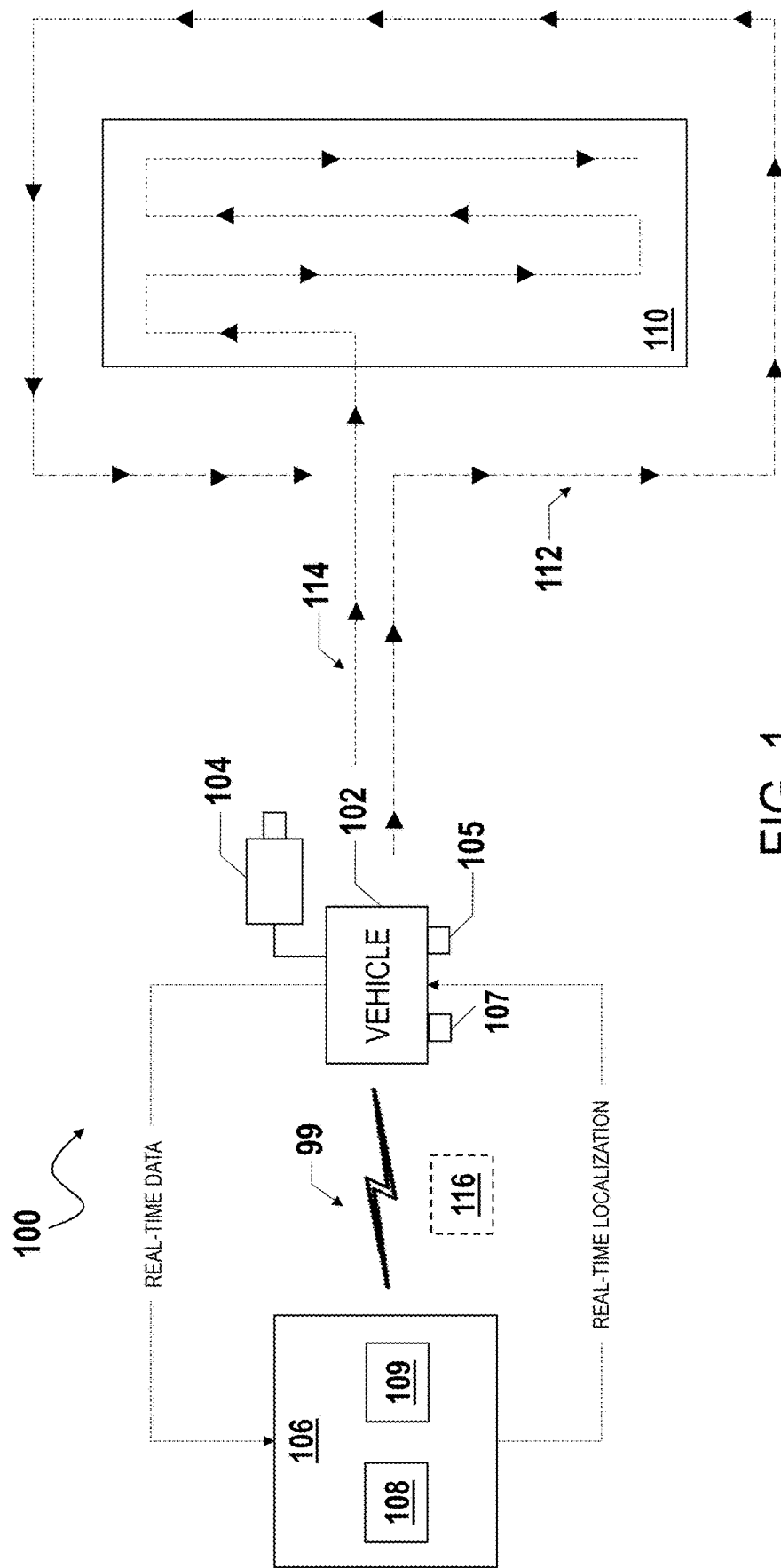


FIG. 1

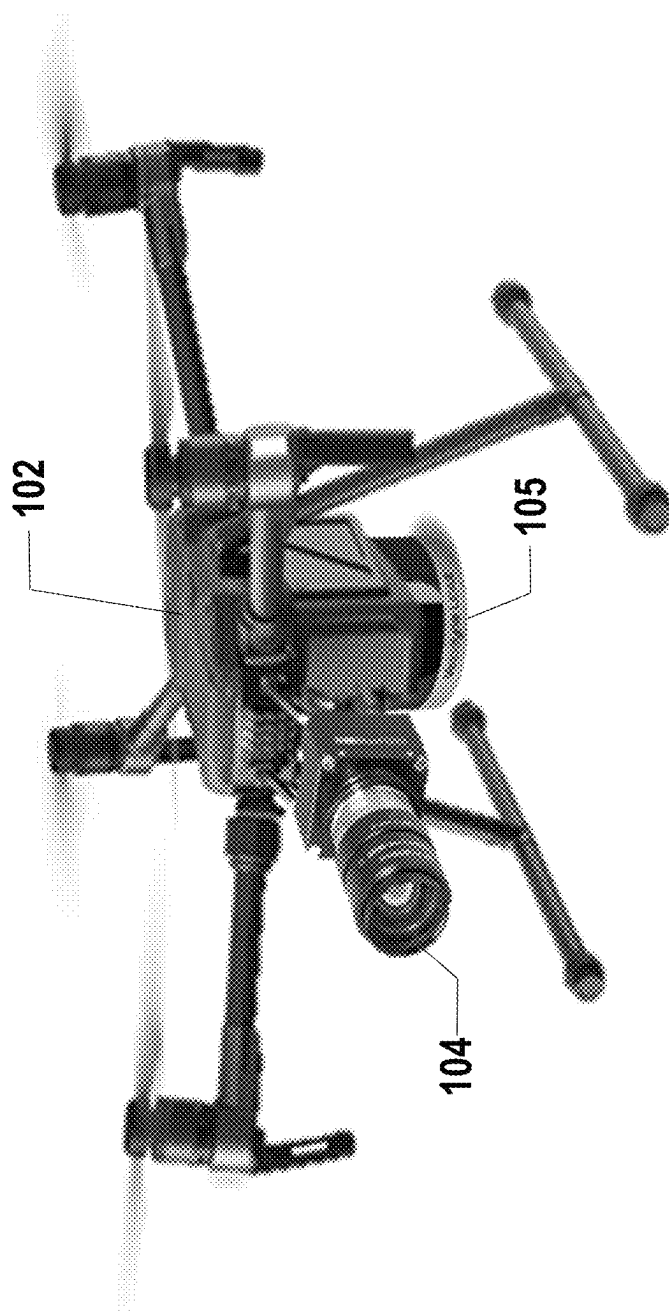
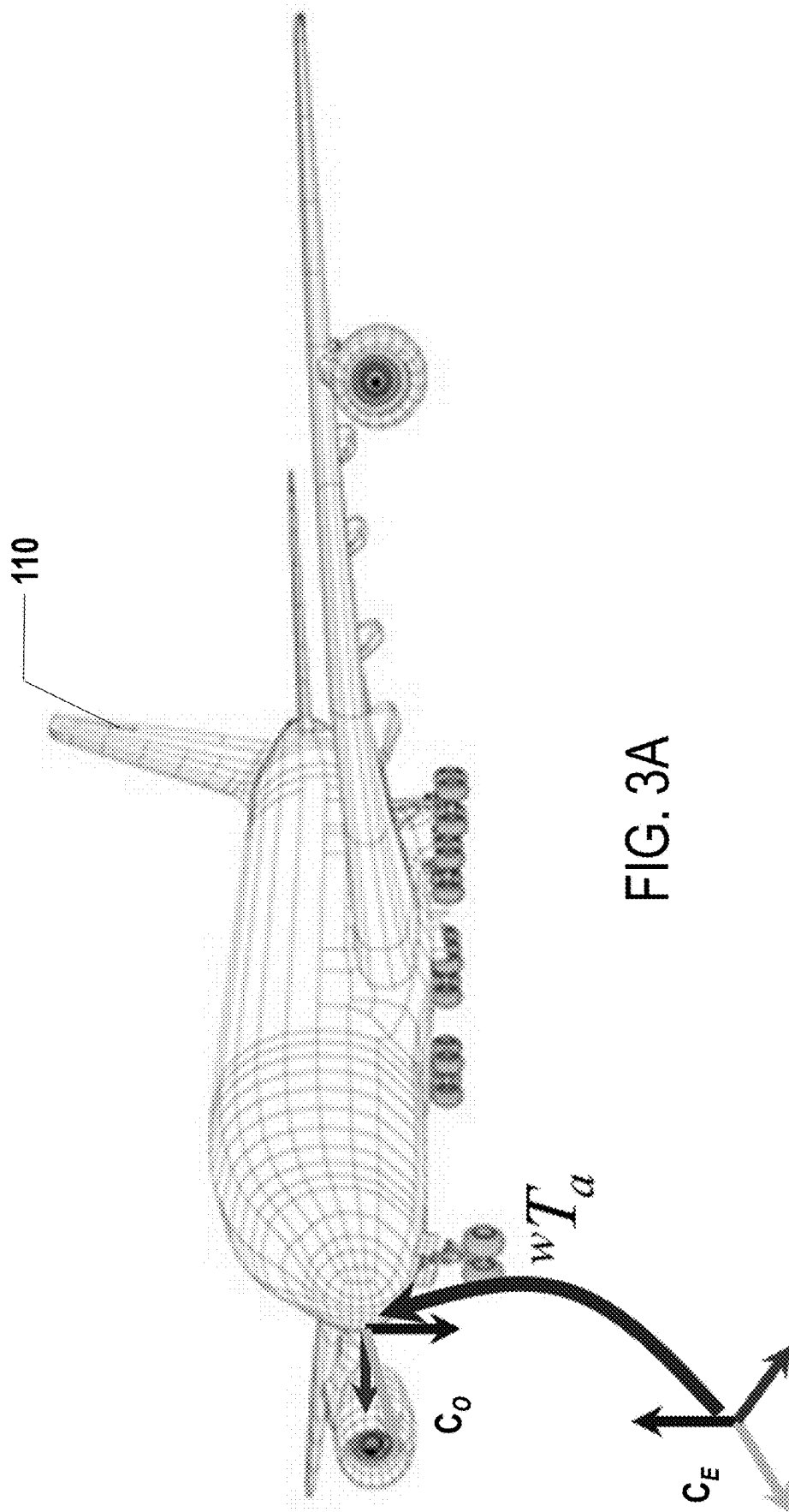
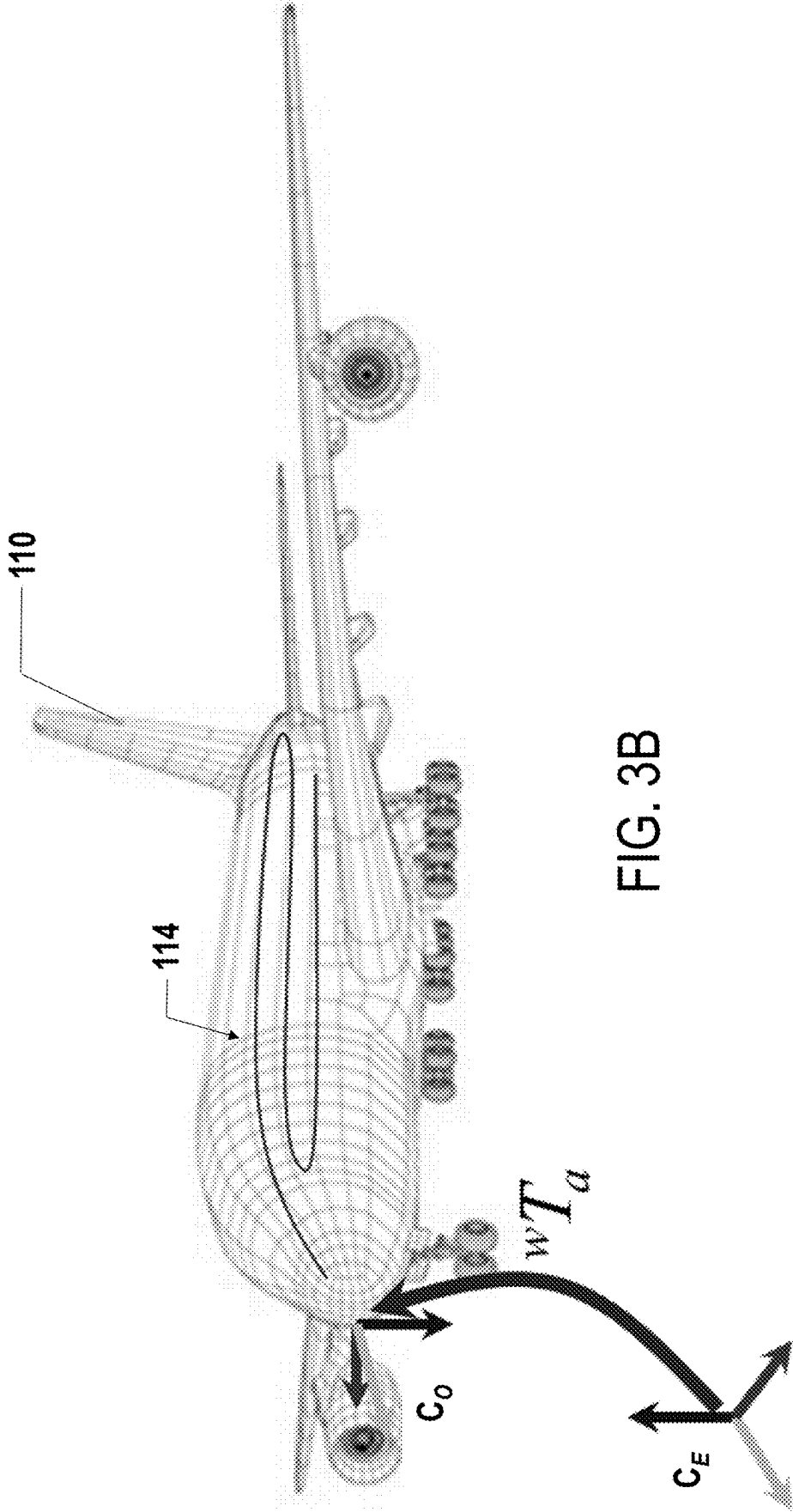


FIG. 2





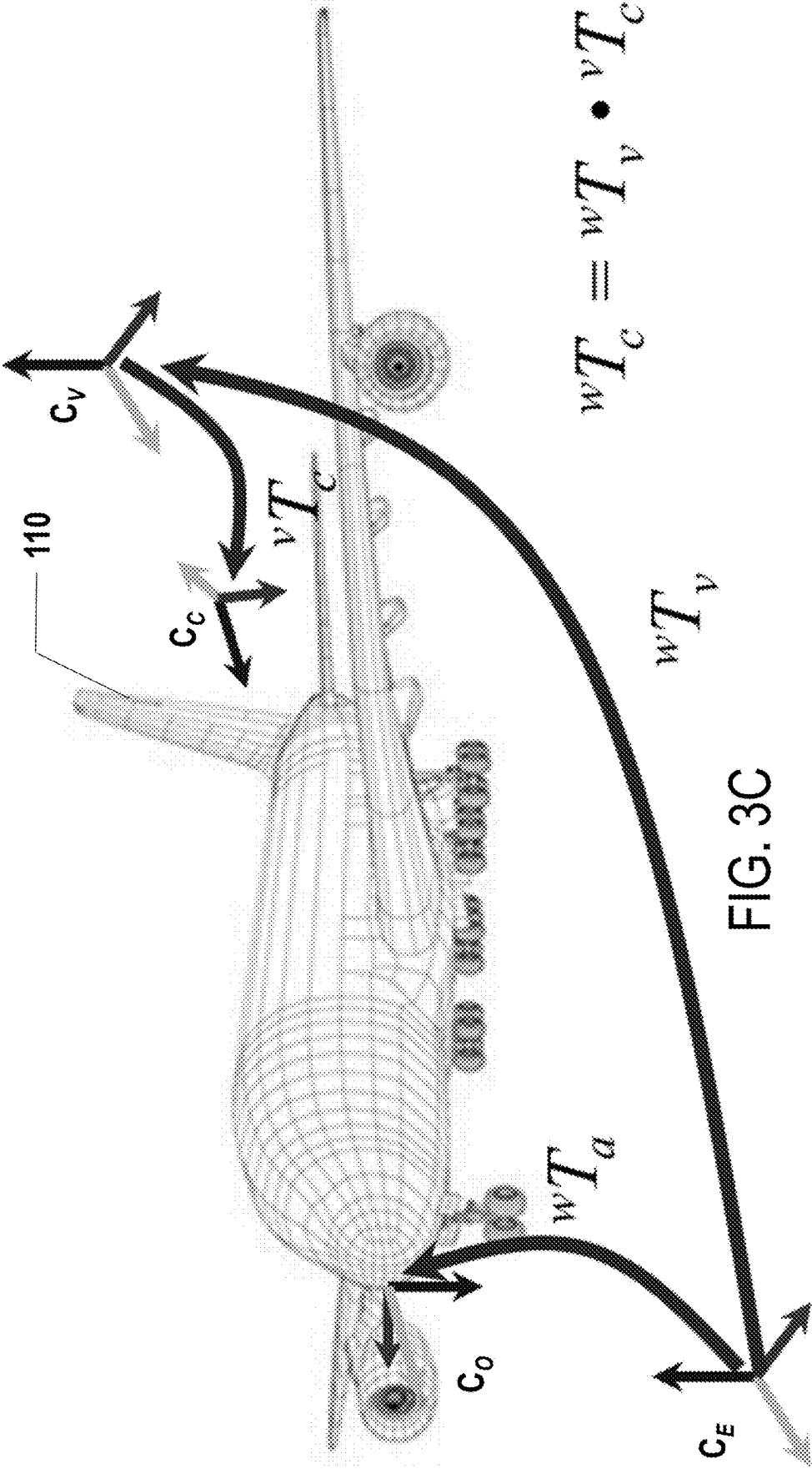


FIG. 3C

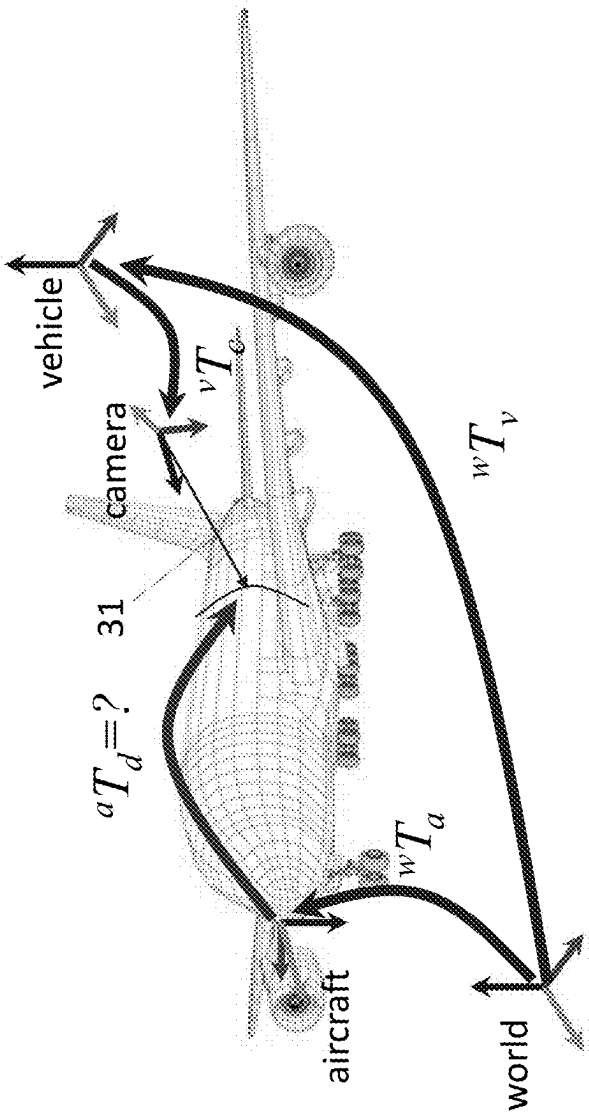
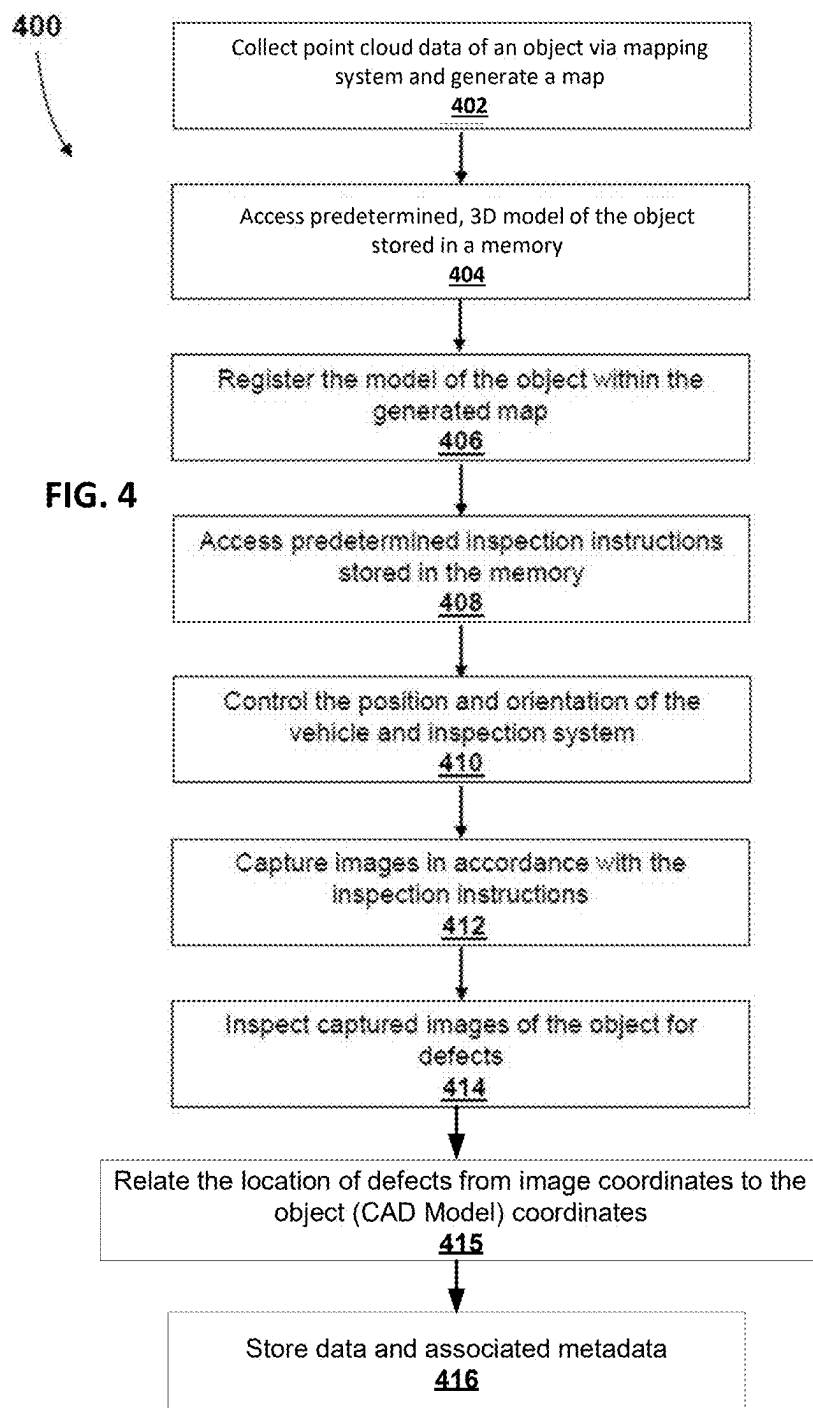


FIG. 3D



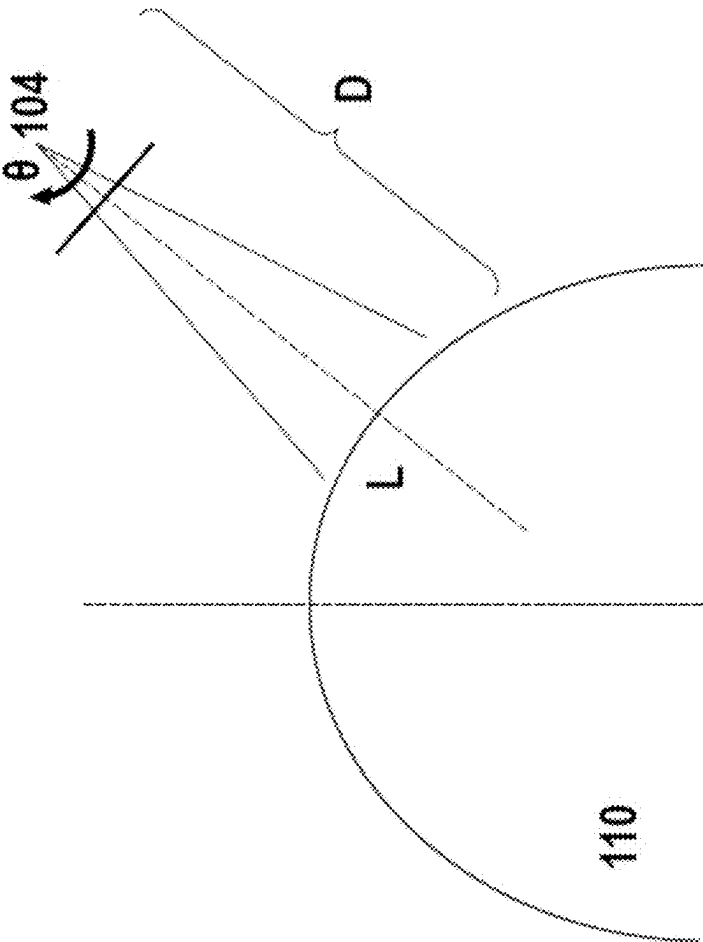


FIG. 5

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VEHICLES, SYSTEMS, AND METHODS FOR THE AUTONOMOUS INSPECTION OF OBJECTS

PRIORITY CLAIM

The present application claims priority to U.S. provisional patent application Ser. No. 63/287,598, filed Dec. 9, 2021, which is incorporated herein by reference in its entirety.

BACKGROUND

Large objects, such as airplanes and architectural structures (e.g., building and bridges) often need inspection. Drones equipped with cameras or other inspection devices and sensors are an emerging way of inspecting such large objects because of their relatively low cost and maneuverability.

SUMMARY

In one general aspect, the present disclosure is directed to an improved method for autonomously inspecting an object, such as an aircraft, a bridge, a building etc. with the autonomous inspection vehicle, such as a drone or unmanned aerial vehicle (UAV). The inspection vehicle includes a localization subsystem, one or more inspection sensors, and a computing system. Optionally the UAV inspection system includes a communication means and a ground station for interaction with a user or for off-board computing. The inspection sensor may include 3D mapping sensor to collect a 3D map (e.g., point cloud) of the object to be inspected relative to a fixed coordinate system. The inspection sensor(s) can characterize the object; i.e., perform the inspection. The computing system includes a memory configured to store a three-dimensional model of the object (e.g., a CAD model) and predetermined inspection instructions. The computing system further includes a processor configured to register the 3D model of the object to the 3D map of the object generated from the 3D mapping sensor. For example, the inspection vehicle can make an initial journey (or the first phase of a single journey) around the object to generate the 3D point cloud of the object. Then the processor can register the 3D CAD model to the 3D point cloud to establish a correspondence of the location of features in the 3D CAD model to the coordinates in the 3D point cloud. The processor can then control the aircraft, in a second journey (or second phase of a single journey) around the object being inspected to perform the inspection in accordance with the inspection instructions and travel path. That is, for example, the aircraft can fly to specific locations required by the inspection instructions to inspect specific features of the object in the 3D model. The inspection system can comprise, for example, a camera, a radar sensor, and/or a thermal imager.

These and other benefits that are realized aspects of the present disclosure will be apparent from the description that follows.

FIGURES

Various aspects of the present disclosure are described herein by way of example in connection with the following figures, wherein:

FIG. 1 illustrates a system configured for the autonomous inspection of an object in accordance with at least one non-limiting aspect of the present disclosure;

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FIG. 2 illustrates a vehicle of the system of FIG. 1 in accordance with at least one non-limiting aspect of the present disclosure;

FIGS. 3A-3D illustrate a model generated by the system of FIG. 1 in accordance with at least one non-limiting aspect of the present disclosure;

FIG. 4 illustrates a method of autonomously inspecting an object in accordance with at least one non-limiting aspect of the present disclosure; and

FIG. 5 illustrates the positioning of an inspection system of the vehicle of FIG. 2 relative to an object to be inspected in accordance with at least one non-limiting aspect of the present disclosure.

DESCRIPTION

The present disclosure is directed, in various aspects, to devices, systems, and methods associated with autonomous vehicles configured to inspect objects. It shall be appreciated that the non-limiting aspects disclosed herein are exclusively presented for illustrative purposes. For example, according to some non-limiting aspects, the vehicle used for inspection can be an unmanned aerial vehicle (UAV) or a drone. However, the present disclosure is not necessarily so limited and other types of inspection vehicles could be used. In that connection, the inspection vehicle may be any suitable type of mobile ground or aerial vehicle, such as, as non-limiting examples, a ground-based mobile robot, a rotor or fixed-wing aircraft, such as a drone, unmanned aerial vehicle, etc. As such, the inspection vehicle can include steering and propulsion systems. For example, the steering/propulsion system for a gas-powered ground vehicle may include a gas-powered engine and the drivetrain. The steering/propulsion system for an electric vehicle may include a battery, an electric motor and the drivetrain. The steering/propulsion system for a rotorcraft (e.g., a drone or other type rotorcraft) can include the rotorcraft's motor-powered rotor(s). The steering/propulsion system for a fixed-wing aircraft can include the engine(s) along with ailerons, elevators, rudders, spoilers, and/or air brakes for steering, for example.

Additionally, the term "autonomous" shall not be limited to unmanned vehicles. For example, a vehicle can be autonomous and still accommodate a human passenger. As used herein, the term "object" when referring to the object being inspected shall also be broadly construed. Although the present disclosure describes non-limiting aspects wherein an autonomous vehicle is configured to inspect an aircraft or an airliner, it shall be appreciated that the vehicles, systems and methods disclosed herein can be similarly implemented to autonomously inspect other types of objects, such as a watercraft or an architectural structure, such as a bridge or building. Accordingly, the term present disclosure contemplates numerous aspects including various vehicles configured to travel substantially without human intervention in at least one mode of operation, and to inspect various objects.

Referring now to FIG. 1, a system **100** including a vehicle **102** configured to autonomously navigate and inspect an object **110** is depicted in accordance with at least one non-limiting aspect of the present disclosure. According to the non-limiting aspect of FIG. 1, the system **100** can include a vehicle **102** that includes one or more inspection sensors **104** for inspecting the object **110**. The inspection sensor **104** can be selected for a specific type of inspection according to user preference and/or intended application. For example, according to the non-limiting aspect of FIG. 1, the inspection sensor **104** can include a camera configured for a visual

inspection of the object **110** and can generate image data (e.g., pictures, videos). The data can be transmitted and processed to characterize the object **110** being inspected. However, it shall be appreciated that the sensor **104** can include any other device suitable for the desired inspection. For example, according to other non-limiting aspects of the present disclosure, the sensor **104** can be configured for radar, sonar, thermal, contact sensing modality, and/or any combination thereof to perform object inspection.

In further reference to FIG. 1, the vehicle **102** can also include a 3D mapping sensor **105** configured to detect three-dimensional shapes within the environment in which the vehicle **102** moves. For example, according to the non-limiting aspect of FIG. 1, the 3D mapping sensor **105** can include a lidar or laser scanner configured to use light in the form of a pulsed laser to measure ranges and/or variable distances between the vehicle **102** and objects within the environment. As such, when the vehicle **102** moves in the environment around the object **110**, the data from the 3D mapping sensor **105** of FIG. 1 can be used to generate a 3D map of the object **110** relative to a coordinate system, e.g., a global coordinate system.

The vehicle **102** includes a localization subsystem **107**. The localization subsystem **107** estimates the position and attitude of the vehicle **102** in the working environment. The localization subsystem uses one or more localization sensors (not shown), such as an inertial measurement unit (IIU), cameras, laser range finders, and/or GPS system. These sensors in combination with signal processing algorithms estimate and track the position and attitude of the vehicle **102** as it moves around the environment. One reference describing and teaching various state-of-the-art localization subsystems for autonomous systems is “Survey on Localization Systems and Algorithms for Unmanned Systems” authored by Shenghai Yuan, Han Wang, and Lihua Xie, published in “Unmanned Systems” Journal, Vol. 9, Number 2, pages 129-163, year 2021, doi:10.1142/S230138502150014X, which is incorporated herein by reference. The knowledge of the vehicle position and attitude afforded by the localization subsystem can be used to reassemble the 3D points detected by the 3D mapping sensor **105** into a coherent 3D point cloud represented in a fixed coordinate system.

The vehicle **102** may also include additional sensors for navigation and obstacle detection and avoidance. In various aspects, output from the inspection sensor(s) **104** could also be used, at various times, for navigation, obstacle detection, and even localization.

Still referring to FIG. 1, the system typically includes a computing system **106**. The computing system **106** could be on-board the vehicle **102** or distributed across both an on-board computing device and a remote, off-board computer system. Whether on-board or distributed on/off-board, the computing system **106** is communicably coupled to the inspection sensor **104** and the 3D mapping sensor **105**, and the localization subsystem **107** via a data link **99**. As used herein, the data link **99** shall include any configuration in which signals can be transmitted between the inspection sensor **104** the computing system **106** or the 3D mapping sensor **105** and the computing system **106**. The data link **99** can include any conventional means of wired and/or wireless communication—both infrastructure and ad hoc. Accordingly, the computing system **106** can be communicably coupled to the inspection sensor **104** and the 3D mapping sensor **105** via a number of different connections, including, but not limited to a local area network (LAN), a coaxial cable, a radiofrequency (RF) transmission, an infra-

red transmission, WiFi®, Bluetooth®, near-field communication (NFC), and/or a cellular connection, amongst others. It shall be appreciated that the data link **99** can include any intermediate systems, subsystems, and/or components positioned between the inspection sensor **104**, the localization sensor **105**, and the computing system **106**.

In further reference to FIG. 1, the computing system **106** can include a processor **108** configured to receive and process signals received from the localization subsystem, the inspection sensor **104** and the 3D mapping sensor **105**, as well as a memory **109** configured to store a predetermined (or pre-established), 3D model of the object **110** (e.g., computer-aided design (CAD) model) to be inspected and predetermined inspection instructions. As such, the vehicle **102** can transmit data from the inspection sensor **104** and the 3D mapping sensor **105** in real-time to the computing system **106**. The processor **108** can: (1) generate a 3D map (e.g., a point cloud) in a coordinate system of the object **110** from the data output by the vehicle’s 3D mapping sensor **105**; (2) marry or align the 3D CAD model of the object **110** to the 3D map; and then (3) control the vehicle to perform the inspection based on the inspection instructions and the knowledge of where the features of the object in the 3D CAD model are from the alignment of step (2). The generation of the 3D map (e.g., step (1) above) and be accomplished with a first journey (or first set of journeys) by the vehicle **102** around the object **110**, with the inspection (step (3)) being performed in a second, subsequent journey (or set of journeys) by the vehicle **102**. In other aspects, all three steps could be performed in one journey: in an initial phase of the journey captures the 3D point cloud data for the object **110**; then aligns the 3D map of the object to a predetermined (e.g., CAD) model of the object that is stored in the memory **109**; and then controls the vehicle **102** to perform the inspection in accordance with the applicable inspection instructions (also stored in the memory) and based on the registration (i.e., alignment) of the predetermined 3D model of the object to the 3D map of the object (which is in a (e.g., global) coordinate frame). Because the map is known in a global coordinate frame, after the registration the predetermined 3D model of the object is localized and known within the same global frame.

As described herein, the system **100** of FIG. 1 can generate the 3D map of the object prior to the inspection. For example, the vehicle **102** of FIG. 1 can be configured to traverse an initial route **112** about the object **110** and, as the vehicle **102** traverses the initial route **112**, the 3D mapping sensor **105** can acquire, for example, 3D point cloud data to generate the 3D map which captures the shape the object **110** in the environment as a 3D point cloud. In a sense, traversing the initial route **112** can be exploratory to generate the 3D map of the object **110**. The initial route **112** is depicted in FIG. 1 as being in one horizontal plane relative to the object **110**, but in other aspects the initial route **112** can include movement in all three directions (x, y and z) relative to the object **110** to collect the necessary point cloud data to generate a sufficient 3D map to carry out the inspection. The processor **108** can generate the 3D map from the point cloud output from the 3D mapping sensor **105**.

In further reference to FIG. 1, upon generating the 3D map, the processor **108** can register the CAD model of the object **110** stored in memory **109** with the coordinate-based 3D map in order to, therefore, register the features in the CAD model with the 3D point cloud of object **110**. For example, the processor **108** can register the CAD model to the 3D map by mapping distinctive geometric features of the CAD model of the object (e.g., wings of an aircraft) to

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corresponding geometric features of the 3D map. An algorithm, such as the iterative closest point (ICP) algorithm, can be used to minimize the errors between the alignments of the various features of the CAD model to the 3D map to ascertain alignment of the 3D model to the 3D map.

Still referring to FIG. 1, the memory 109 can be further configured to store instructions associated with a desired inspection of the object 110. The memory 109 can be configured to store a predetermined inspection route 114 about the object 110. The inspection route 114 can include instructions to capture images of certain features of the object 110 in the 3D (CAD) model from certain directions. For example, FIG. 5 depicts an a non-limiting aspect wherein the object 110 is an aircraft fuselage to be inspected and the inspection route 114 instructs the vehicle 102 to capture an image of a specific location L on the fuselage from a predetermined distance D (as a non-limiting example, 1.3 to 2.3 m) at a predetermined angle θ relative to the aircraft. The specific location L on the aircraft 110 can be specified relative to the 3D CAD model of the aircraft 110. After the alignment of the 3D CAD model to the 3D point cloud of the object 110 and the surrounding scene (i.e., the scene around the object), the location L becomes relative to the 3D map of the real location of the object 110 in the environment. The processor 108 can be configured to determine: (i) the 3D coordinates of the features of the object that need to be imaged from the alignment of the 3D model to the 3D map, (ii) the real-time, continuously-updated position and orientation (also known as “pose”) of the vehicle during the inspection journey, and (iii) the orientation of the camera’s field of view relative to the vehicle’s pose. The processor 108 can control the operation and movement of the vehicle 102 based, at least in part, on the determined information, such that the vehicle 102 travels to the specific location L, from the predetermined distance D from the object 110, and orients itself at the predetermined angle θ , such that the inspection system 104 is appropriately oriented according to inspection instructions. Accordingly, the vehicle 102 can capture a desired image of the object 110 for inspection.

The inspection system (e.g. camera) 104 can be fixedly positioned on the vehicle, such that the orientation of the camera relative to the vehicle is constant. In other aspects, the inspection system 104 can be movable in one or more directions, and the processor 108 can continually track a direction of the inspection system 104 relative to the vehicle 102. The vehicle 102 can include other on-board detection and avoidance systems which it can use to navigate around other objects as it traverses the initial journey to generate the 3D map and performs an inspection of the primary object 110.

As the vehicle 102 traverses the inspection route 114, the inspection sensor 104 can collect and transmit data associated with a condition of the object 110 to the processor 108. The processor 108 can be further configured to detect defects in the object 110 based, at least in part, on the data collected by the inspection system 104. For example, according to some non-limiting aspects, the processor 108 can detect defects in the object 110 based, at least in part, on a comparison of an image captured by the inspection sensor 104 to an image of the object 110 stored in the memory 109. According to other non-limiting aspects, the processor 108 can implement a machine learning algorithm, such as a deep neural network, that is trained to detect defects in the object 110 based on input data from the inspection system 104. Based on the time-stamped location of the inspection system 104 or vehicle 102 when an image captured by the inspec-

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tion image reveals a defect, the processor 108 can determine a location of the defect on the object relative to the 3D model and/or the 3D map.

Still referring to FIG. 1, according to some non-limiting aspects, the computing system 106 can further include a user interface 116 configured to illustrate the determined location of the detected defect(s) of the object 110 and/or image data collected by the inspection sensor 104. The user interface 116 can be on-board or off-board (remote from) the vehicle 102. The user interface 116 can be communicably coupled to the computing system 106 via the data link 99, can include a display, and can be implemented with a separate computer and/or a mobile device.

FIG. 2, depicts the vehicle 102 of FIG. 1 in accordance with at least one non-limiting aspect. According to the non-limiting aspect of FIG. 2, the inspection vehicle 102 can be an autonomous drone, or unmanned aerial vehicle (UAV), configured to travel by air. The vehicle 102 can include the inspection sensor 104 configured to perform the inspection of the object 110 (e.g., a camera to capture images of the object 110).

In further reference to FIG. 2, the vehicle 102 can further include the 3D mapping sensor 105, which can be a lidar or laser scanner. For example, the lidar sensing system 105 senses objects (e.g., surfaces of the object 110) in the vicinity of the vehicle 102 by measuring the distances from the 3D mapping sensor 105 (and hence the vehicle 102) to the objects by illuminating the surfaces/objects with laser light. In one aspect, the lidar laser system can include several channels (e.g., 16 or 32 laser beams) that scan a 360° horizontal field of view, with about a 30° to 40° vertical field of view, that scans 5 to 20 times per second (for example), thereby collecting hundreds of thousands (e.g., 300,000) lidar data points per second, with a measurement range of about 100 meters and an accuracy of about 2 cm or less. The lidar sensing system 105 can be implemented with a Velodyne® HDL-32E model lidar sensor, for example, or any other suitable lidar scanner, assuming it has a suitable range, bandwidth, and/or field of view for the intended application.

FIGS. 3A-3D illustrate how the inspection of the object 110 may be carried out according to several non-limiting aspects of the present disclosure. According to the aspects of FIGS. 3A-3D, the object 110 can be an aircraft. As shown in FIG. 3A, the 3D map of the object (e.g., the point cloud of object 110) can be expressed in terms of a global 3D coordinate system C_w . For example, the 3D mapping system chooses the origin of the global 3D coordinate system C_w , and generates a coherent point cloud descriptive of the shape of the object 110. The alignment of the 3D (CAD) model to the 3D map, performed by the processor, discovered a rigid six degrees of freedom (6DOF) transform wT_A , by which the 3D CAD model reference frame C_o is placed within coordinates in the C_e coordinate system. FIG. 3B illustrates how an example travel path 114 that is originally defined in the 3D CAD coordinate system is aligned with the real 3D map of the object 110 after applying the 6DOF transform wT_A . According to the non-limiting aspect of FIG. 3B, the vehicle 102 can fly back and forth along the fuselage at different height levels, or any arbitrary flight path predefined in the memory 109 according to the need of a particular inspection task.

As mentioned herein, the inspection instructions stored in the memory 109 can include instructions to capture images of specific locations on the object 110, at certain distances from the object 110, and at certain angles relative to the object 110. For example, FIG. 5 illustrates one such location, distance, and angle that can be included in the inspection

instructions. The inspection vehicle **102** motion controller in combination with information from the localization subsystem positions the vehicle **102** and the inspection sensor **104** in the correct location and with the correct orientation relative the object **110** to capture the images required by the inspection instructions. FIG. **3C** shows the localization system reporting 6DOF transform ${}^W T_v$, the pose of the vehicle's **102** reference frame C_v in the global reference frame C_w . The pose tracking system of the inspection sensor **104** determines 6DOF transform ${}^V T_c$, the pose of the inspection sensor's **104** reference frame C_c in the vehicle's reference frame C_v .

FIG. **3D** shows a particular sensing ray **31** of the inspection sensor **104** (e.g., camera) whose direction is known in the reference frame C_c by the virtue of the inspection sensor **104** construction and calibration. For example, when the inspection sensor is a camera, the inspection ray **31** can be thought of as a viewing ray of one pixel from the image sensor. This pixel may be where a defect is discovered via image analysis. It is of interest to determine a location where this defect is on the object **110**. FIG. **3D** shows how this location can be found. The location of the intersection of the sensing ray **31** with the surface of the object **110** can be determined by compounding all the known rigid transforms (and their inverses where appropriate) ${}^W T_d$, ${}^W T_v$, ${}^V T_c$, and the direction of the sensing ray **31**. For those skilled in art the compounding of these transforms is trivial yielding the location of the defect within the 3D CAD model of the aircraft. This location directly guides a maintenance personnel to the location of the defect on the aircraft, or this location is logged into the maintenance system.

Referring now to FIG. **4**, a method **400** of autonomously inspecting the object **110** with the vehicle **102** is depicted in accordance with at least one non-limiting aspect of the present disclosure. According to the non-limiting aspect of FIG. **4**, the method **400** can include, at step **402**, collecting 3D point cloud data of the object **110** with the 3D mapping system **105** of the vehicle **102** and generating the 3D map (e.g., point cloud) of the object from the point cloud data. As previously discussed, the collection of the 3D point cloud data can be performed during an initial journey of the vehicle **102** through an environment within which the object **110** is positioned. The initial journey can be considered a first (or initial) movement stage (or phase) of the vehicle around the object. The vehicle may move around the vehicle one or more times in the first movement stage/phase. In the case of an aerial inspection vehicle **102**, the vehicle **102** could land one or more times during the course of the first movement stage/phase. The lidar point cloud data can be time-stamped and the 3D location of the vehicle is known continuously as the point cloud data are captured. Thus, the lidar point cloud for the object **110** can be registered to a 3D global coordinate system (e.g., CE in FIGS. **3A-3C**). The method can further include, at step **404**, accessing a 3D model (e.g., a CAD model) of the object **110** stored in the memory **109** of the computing system **106**, of the vehicle **102**.

According to the method **400** of FIG. **4**, at step **406**, the computing system **106** registers the 3D model of the object to the 3D map generated at step **404**. At step **408**, predetermined inspection instructions stored in the memory **10** of the computing system **106** can then be accessed. At step **410**, the processor **108** can control the position and orientation of the vehicle **102**, and correspondingly the position and orientation of the inspection system **104** on the vehicle **102**, as the vehicle **102** moves around the object **110** for the inspection so that, at step **412**, the inspection system **104** can capture

the images of the object **110** in accordance with the inspection instructions. The movement of the vehicle **102** around the object **110** at step **410** can be considered a second movement stage (or phase) of the vehicle that follows in time the first movement stage/phase. At step **414**, the computing system **106** can inspect the images captured by the inspection system **104** to detect defects (or lack of defects) in the captured images of the object **110**. At step **415** the computing system **106** relates the image coordinates of the detected defects to the 3D coordinates on the object **110** CAD model where the detected defect is. At step **416** the inspection data and metadata related to detected defects and their coordinates are stored in a database for further use, historical record, and maintenance.

In various aspects, the vehicle **102** may land after step **402** and before step **410** (e.g., between in time the first and second movement stages of the vehicle) so that the batteries of the vehicle **102** can be conserved and/or recharged. Then the vehicle **102** can take off again to inspect the vehicle at steps **410** and **412**. In other aspects, steps **402** to **412** could be completed in one flight by the vehicle **102**. The defect detection at step **414** could be performed by an on- or off-board computer of the computing system **106**. When an off-board computer performs the defect detection, the images captured by the inspection system **104** can be uploaded to the off-board computer system after steps **410**, **412**, or, in other aspects, could be transmitted during the flight to the off-board computer for inspection. Also, the registering/aligning of the 3D model to the 3D map at step **404** could be performed by an on- or off-board computer of the computer system **106**. When the registration/alignment is performed off-line, the results can be uploaded to the vehicle **102** so that the vehicle's processor **108** can control the flight path of the vehicle for the inspection.

The processor **108** can also include a controller for controlling the steering, propulsion and/or braking systems of the vehicle **102** based on navigation control commands generated by the processor **108**. The processor **108**, whether on-board the vehicle **102**, off-board the vehicle **102**, or distributed between the vehicle **102** and a remote computer system, may include one or more processors (e.g., CPUs or GPUs), primary data storage or memory (i.e., memory that is directly accessible to the CPUs/GPUs, such as RAM, ROM, registers, cache memory), and secondary data storage (i.e., data storage that is not directly accessible by the CPUs/GPUs, such as HDDs, flash, SSDs, etc.). The computer system may also include nearline and/or off-line storage. The processor(s) **108** and the computer system may be programmed to perform the functions described herein with software that is stored in the primary, secondary, nearline and/or off-line data storage, as the case may be, and executed by the processor(s) **108**. The computer software may be implemented using any suitable computer programming language such as NET, C, C++, JavaScript, Python, Ruby, Lua, and Perl, and using conventional, functional, or object-oriented techniques. Programming languages for computer software and other computer-implemented instructions may be translated into machine language by a compiler or an assembler before execution and/or may be translated directly at run time by an interpreter.

In one general aspect, therefore, embodiments of the present invention are directed to systems and methods for inspecting an object, such as for defects in the object. The system, in various embodiments, comprises a vehicle for moving around the object to inspect the object, where the vehicle comprises: a 3D mapping sensor system for collecting time-stamped first data of an environment around the

vehicle as the vehicle moves around the object; a localization sensor system for collecting time-stamped second data indicative of position and attitude of the vehicle as the vehicle moves around the object; and an inspection sensor system for collecting time-stamped third data descriptive of the object as the vehicle moves around the object, where the third data are for detecting a possible defect with the object. The system also comprises a computer system that comprises: a memory for storing a predetermined 3D model of the object; and a processor programmed to operate vehicle motion in two stages around the object. The two stages comprise a first motion stage and a second stage, the second stage being subsequent to the first stage. From the first motion stage of the vehicle around the object, the computer system is configured to: generate a 3D map of the environment based on the first data collected by the 3D mapping sensor system of the vehicle and the second data collected by the localization sensor system of the vehicle; and register the predetermined 3D model of the object within the 3D map, thereby generating a registration identifying where the predetermined 3D model is located within the 3D map.

From the second motion stage of the vehicle around the object, the computer system is configured to: control the position and orientation of the vehicle as the vehicle moves around the object based on, at least, predetermined inspection instructions for the vehicle, where the predetermined inspection instructions are stored in the memory; and inspect the third data collected by the inspection sensor system to identify one of more defects in the object. For each of the one or more defects identified in the object, the computer system is configured to relate a location of the defect on the object from coordinates for the defect from the third data to coordinates of the predetermined 3D model of the object based on the registration of the predetermined 3D model of the object with the 3D map.

A method for inspecting an object can comprise the step of operating a vehicle motion in two stages around the object, the two stages comprising a first motion stage and a second stage, the second stage being subsequent to the first stage. In the first motion stage of a vehicle around the object, the method comprises: collecting, by a 3D mapping sensor system of the vehicle, time-stamped first data of an environment around the vehicle as the vehicle moves around the object; collecting, by a localization sensor system of the vehicle, collecting time-stamped second data indicative of a position and attitude of the vehicle as the vehicle moves around the vehicle; generating, by a computer system, a 3D map of the environment based on the first data collected by the 3D mapping sensor system of the vehicle and the second data collected by the localization sensor system of the vehicle; and generating, by the computer system, a registration of a predetermined 3D model of the object with the 3D map, where the predetermined 3D model of the object is stored in a memory of the computer system;

In the second motion stage of the vehicle around the object, the method comprises the steps of: collecting, by an inspection sensor system of the vehicle, time-stamped third data of the object as the vehicle moves around the object, where collecting the third data comprises controlling position and orientation of the vehicle for the second motion stage of the vehicle according to predetermined inspection instructions for the vehicle, wherein the predetermined inspection instructions are stored in the memory of the computer system; and inspecting, by the computer system, the third data collected by the inspection sensor system to identify one of more defects in the object. For each of the one or more defects identified in the object, the method

comprises the steps of: relating, by the computer system, a location of the defect on the object from coordinates for the defect from the third data to coordinates of the predetermined 3D model of the object based on the registration of the predetermined 3D model of the object within the 3D map.

In various implementations, the vehicle comprises an autonomous aerial vehicle, such as a drone. The object being inspected may be an airplane. The inspection sensor system may comprise a camera such that the third data comprises image data. The 3D mapping sensor system may comprise a lidar sensor. The localization sensor system may comprise at least one of an inertial measurement unit, a laser range finder, a camera, and a GPS receiver. The predetermined 3D model of the object may comprise a CAD model of the object. The 3D map may comprise a point cloud of a scene in which the inspection system operates. The computer system may be programmed to register the predetermined 3D model to the 3D map by registering one or more distinctive 3D features of the predetermined 3D model of the object to corresponding distinctive 3D features in the 3D map.

In various implementations, the computer system is programmed to register the predetermined 3D model to the 3D map by adjusting an alignment between the predetermined 3D model and the 3D map by minimizing a distance between the one or more distinctive features of the predetermined 3D model of the object to the corresponding distinctive features in the 3D map.

The computer system may be on-board the vehicle or it could be off-board the vehicle, in which case the vehicle can communicate with the computer system via a wireless data link.

In various implementations, the computer system is programmed to examine the third data collected by the inspection sensor system to identify one of more defects in the object using a machine learning algorithm that is trained to detect defects in the object. Also, the possible defect can comprise a possible flaw in the object, a possible anomaly in the object, and/or a possible exception in the object (e.g., something out of nominal).

The vehicles, systems and methods disclosed herein shall not be limited to the above-described configurations. Rather, the specific configurations described and depicted herein are exclusively presented for illustrative purposes. It shall be appreciated that any of the features described herein can be reallocated based on user preference and/or intended application. For example, although some non-limiting aspects may describe a vehicle with an on-board computing system, it shall be appreciated that, according to other non-limiting aspects, the computing system—or any of its components—can be consolidated, partitioned, repositioned, and/or reconfigured to achieve the same functional effect. For example, according to other non-limiting aspects, the computing system—or any of its components—can be remotely located relative to the vehicle. In short, the vehicles, systems, and methods disclosed herein shall not be limited to any particular system architecture.

The aspects presented herein are intended to illustrate potential and specific implementations of the present disclosure. It can be appreciated that the examples are intended primarily for purposes of illustration of the invention for those skilled in the art. No particular aspect or aspects of the examples are necessarily intended to limit the scope of the present disclosure. Further, it is to be understood that the figures and descriptions of the present disclosure have been simplified to illustrate elements that are relevant for a clear

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understanding of the present disclosure, while eliminating, for purposes of clarity, other elements. While various aspects have been described herein, it should be apparent that various modifications, alterations, and adaptations to those aspects may occur to persons skilled in the art with attainment of at least some of the advantages. The disclosed aspects are therefore intended to include all such modifications, alterations, and adaptations without departing from the scope of the aspects as set forth herein.

What is claimed is:

1. A system for inspecting an object, the system comprising:
 - a vehicle for moving around the object to inspect the object, the vehicle comprising:
 - a 3D mapping sensor system for collecting time-stamped first data of an environment around the vehicle as the vehicle moves around the object;
 - a localization sensor system for collecting time-stamped second data indicative of position and attitude of the vehicle as the vehicle moves around the object; and
 - an inspection sensor system for collecting time-stamped third data descriptive of the object as the vehicle moves around the object, wherein the third data are for detecting a possible defect with the object; and
 - a computer system, wherein the computer system:
 - comprises a memory for storing a predetermined 3D model of the object; and
 - a processor programmed to operate vehicle motion of the vehicle in two stages around the object, wherein the two stages comprise a first motion stage and a second stage, the second stage being subsequent to the first stage, wherein:
 - from the first motion stage of the vehicle around the object, the processor is configured to:
 - generate a 3D map of the environment based on the first data collected by the 3D mapping sensor system of the vehicle and the second data collected by the localization sensor system of the vehicle; and
 - register the predetermined 3D model of the object within the 3D map, thereby generating a registration identifying where the predetermined 3D model is located within the 3D map; and
 - from the second motion stage of the vehicle around the object, the processor is configured to:
 - control position and orientation of the vehicle as the vehicle moves around the object based on, at least, predetermined inspection instructions for the vehicle, wherein the predetermined inspection instructions are stored in the memory;
 - inspect the third data collected by the inspection sensor system to identify one of more defects in the object; and
 - for each of the one or more defects identified in the object, relate a location of the defect on the object from coordinates for the defect from the third data to coordinates of the predetermined 3D model of the object based on the registration of the predetermined 3D model of the object with the 3D map.
2. The system of claim 1, wherein the vehicle comprises an autonomous aerial vehicle.
3. The system of claim 2, wherein the autonomous aerial vehicle comprises a drone.

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4. The system of claim 3, wherein the object comprises an airplane.

5. The system of claim 1, wherein the inspection sensor system comprises a camera and wherein the third data comprises image data.

6. The system of claim 1, wherein the 3D mapping sensor system comprises a lidar sensor.

7. The system of claim 6, wherein the inspection sensor system comprises a camera and wherein the third data comprises image data.

8. The system of claim 7, wherein the localization sensor system comprises at least one of an inertial measurement unit, a laser range finder, a camera, and a GPS receiver.

9. The system of claim 8, wherein the predetermined 3D model of the object comprises a CAD model of the object.

10. The system of claim 9, wherein the 3D map comprises a point cloud of a scene in which the inspection system operates.

11. The system of claim 1, wherein the computer system is programmed to register the predetermined 3D model to the 3D map by registering one or more distinctive 3D features of the predetermined 3D model of the object to corresponding distinctive 3D features in the 3D map.

12. The system of claim 11, wherein the computer system is programmed to register the predetermined 3D model to the 3D map by adjusting an alignment between the predetermined 3D model and the 3D map by minimizing a distance between the one or more distinctive features of the predetermined 3D model of the object to the corresponding distinctive features in the 3D map.

13. The system of claim 1, wherein the computer system is on-board the vehicle.

14. The system of claim 1, wherein:

the computer system is off-board the vehicle; and
the vehicle communicates with the computer system via a wireless data link.

15. The system of claim 1, wherein the computer system is programmed to examine the third data collected by the inspection sensor system to identify one of more defects in the object using a machine learning algorithm that is trained to detect defects in the object.

16. The system of claim 1, wherein:

the vehicle comprises an autonomous aerial vehicle;
the inspection sensor system comprises a camera and wherein the third data comprises image data;
the 3D mapping sensor system comprises a lidar sensor;
the localization sensor system comprises at least one of an inertial measurement unit, a laser range finder, a camera and a GPS receiver;

the predetermined 3D model of the object comprises a CAD model of the object;

the 3D map comprises a point cloud; and

the computer system is programmed to generate the registration by, for each of one or more distinctive features of the predetermined 3D model of the object, mapping the distinctive feature to a corresponding geometric feature of 3D map by minimizing an alignment of the one or more distinctive features of the predetermined 3D model of the object to the corresponding geometric features of 3D map.

17. The system of claim 16, wherein the computer system is on-board the vehicle.

18. The system of claim 1, wherein each of the one or more defects identified in the object comprises a defect selected from the group consisting of a flaw in the object, an anomaly in the object, and an exception in the object.

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19. A method for inspecting an object, the method comprising:

operating a vehicle motion in two stages around the object, the two stages comprising a first motion stage and a second stage, the second stage being subsequent to the first stage;

in the first motion stage of a vehicle around the object:

collecting, by a 3D mapping sensor system of the vehicle, time-stamped first data of an environment around the vehicle as the vehicle moves around the object; and

collecting, by a localization sensor system of the vehicle, collecting time-stamped second data indicative of a position and attitude of the vehicle as the vehicle moves around the vehicle;

generating, by a computer system, a 3D map of the environment based on the first data collected by the 3D mapping sensor system of the vehicle and the second data collected by the localization sensor system of the vehicle;

A generating, by the computer system, a registration of a predetermined 3D model of the object located within the 3D map, wherein the predetermined 3D model of the object is stored in a memory of the computer system;

in the second motion stage of the vehicle around the object, collecting, by an inspection sensor system of the vehicle, time-stamped third data of the object as the vehicle moves around the object, wherein collecting the third data comprises controlling position and orientation of the vehicle for the second motion stage of the vehicle according to predetermined inspection instructions for the vehicle, wherein the predetermined inspection instructions are stored in the memory of the computer system;

inspecting, by the computer system, the third data collected by the inspection sensor system to identify one of more defects in the object; and

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for each of the one or more defects identified in the object, relating, by the computer system, a location of the defect on the object from coordinates for the defect from the third data to coordinates of the predetermined 3D model of the object based on the registration of the predetermined 3D model of the object within the 3D map.

20. The method of claim 19, wherein:

the vehicle comprises an autonomous aerial vehicle;

the inspection sensor system comprises a camera wherein the third data comprises image data;

the 3D mapping sensor system comprises a lidar sensor;

the localization sensor system comprises at least one of an inertial measurement unit, a lidar, a camera, and a GPS receiver;

the predetermined 3D model of the object comprises a CAD model of the object;

the 3D map comprises a point cloud of a scene in which the inspection system operates; and

registering each of one or more distinctive features of the predetermined 3D model of the object, to the distinctive feature to one or more corresponding geometric features in the 3D map by minimizing an alignment error of the one or more distinctive features of the predetermined 3D model of the object to the corresponding geometric features of 3D map.

21. The method of claim 19, wherein the computer system is on-board the vehicle.

22. The method of claim 19, further comprising recharging a rechargeable battery of the vehicle between the first and second motion stages of the vehicle.

23. The method of claim 19, wherein:

the vehicle comprises an aerial vehicle; and

the method comprises completing the first and second motion stages of the aerial vehicle without landing the aerial vehicle until the second motion stage is completed.

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